

SAION KUMAR ROY

611 EXP, 815 Columbus Ave, Boston, MA 02118

Postdoctoral Researcher, Northeastern University

+1(217)-898-2084 ◊ sai.roy@northeastern.edu ◊ 

RESEARCH INTERESTS

My research aims to develop system-on-chip (SoC) co-design methodologies for energy-efficient machine learning accelerators, with security and privacy treated as first-class design constraints alongside performance and implementation cost. My work explores novel architectures, including in-memory computing, and investigates how algorithm mapping, microarchitecture, and circuit implementation choices in silicon shape both energy efficiency and the hardware attack surface. The objective is to orchestrate a unified design paradigm spanning ML algorithms, accelerator architectures, and end-to-end design flows, enabling the secure and efficient deployment of machine learning systems on advanced computing platforms.

WORK EXPERIENCE

Northeastern University

Postdoctoral Researcher, Electrical and Computer Engineering

Advisor: Professor Yunsi Fei

Sept 2024 - Present

EDUCATION

University of Illinois at Urbana-Champaign

PhD, Electrical and Computer Engineering

Advisor: Professor Naresh Shanbhag

Aug 2018 - Aug 2024

GPA: 3.88/4.0

Indian Institute of Technology Kharagpur

Bachelor and Masters of Technology

Electronics and Electrical Communication Engineering

July 2013 - July 2018

GPA: 9.26/10

AWARDS AND ACHIEVEMENTS

- Oral Presentation Award, CSL Student Conference, Security and Privacy Session, 2024.
- Best Speaker Award, CSL Student Conference, Circuits and Systems Session, 2023.
- ISCAS Student Participation Grant Winner, 2022.
- Deutscher Akademischer Austauschdienst (DAAD) WISE Fellow, 2017.

PUBLICATIONS

- T. Xu, **S. K. Roy**, Ruyi Ding, A. A. Ding, and Y. Fei, “SLAC: Access-Driven CPU-to-GPU Side-channel Attacks via System-Level Cache on Apple Silicon”, **ACM SIGSAC Conference on Computer and Communications Security (CCS)**, 2026 (to appear).
- **S. K. Roy**, Yufei Wang, A. A. Ding, and Y. Fei, “ROBBIN: Rowhammer-Based Backdoor Injection during Inference”, **IEEE/ACM International Conference on Computer-Aided Design (ICCAD)**, 2026 (to appear).
- **S. K. Roy**, and N. Shanbhag, “Comprehending the Energy-Accuracy-Security Trade-offs in Embedded Non-Volatile In-Memory Computing Architectures”, **IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)**, 2026.
- **S. K. Roy**, Z. Zhang, A. A. Ding, and Y. Fei, “Attack from Shadows: Unsupervised Side-channel Transfer Learning across Devices and Modalities”, **21st ACM Asia Conference on Computer and Communications Security (AsiaCCS)**, 2026.
- **S. K. Roy**, and N. Shanbhag, “The Energy-Accuracy-Security Trade-off in Resistive In-memory Architectures”, **IEEE International Electron Devices Meeting (IEDM)**, 2024.
- **S. K. Roy**, and N. Shanbhag, “On the Security Vulnerabilities of MRAM-based In-Memory Computing Architectures against Model Extraction Attacks”, **IEEE/ACM International Conference on Computer-Aided Design (ICCAD)**, 2024.
- **S. K. Roy**, H.-M. Ou, M. Ahmed, P. Deaville, B. Zhang, N. Verma, P. Hanumolu, and N. Shanbhag, “Compute SNDR-boosted 22nm MRAM-based In-memory Computing Macro using Statistical Error Compensation”, **IEEE Journal of Solid-State Circuits (JSSC)**, 2024.

- **S. K. Roy**, and N. Shanbhag, “Energy-Accuracy Trade-offs for Resistive In-Memory Computing Architectures”, **IEEE Journal on Exploratory Solid-State Computational Devices and Circuits (JxCDC)**, 2024.
- **S. K. Roy**, H.-M. Ou, M. Ahmed, P. Deaville, B. Zhang, N. Verma, P. Hanumolu, and N. Shanbhag, “Compute SNR-boosted 22nm MRAM-based In-memory Computing Macro using Statistical Error Compensation”, **IEEE European Solid-State Circuits Conference (ESSCIRC)**, 2023.
- N. Shanbhag and **S. K. Roy**, “Benchmarking In-memory Computing Architectures”, **IEEE Open Journal of the Solid-State Circuits Society (OJ-SSCS)**, 2022.
- N. Shanbhag and **S. K. Roy**, “Comprehending In-memory Computing Trends via Proper Benchmarking”, **IEEE Custom Integrated Circuits Conference (CICC)**, 2022.
- **S. K. Roy**, A. Patil, and N. Shanbhag, “Fundamental Limits on the Computational Accuracy of Resistive Crossbar-based In-memory Architectures”, **IEEE International Symposium on Circuits and Systems (ISCAS)**, 2022.

INVITED TALKS

- “Attack from Shadows: Unsupervised Side-channel Transfer Learning across Devices and Modalities”, **ACM AsiaCCS**, at Bangalore, India, 2026.
- “Energy-Accuracy-Security Trade-offs in Resistive In-Memory Computing Architectures”, **New England Hardware Security Day**, at MIT, Boston, Massachusetts, USA 2025.
- “Energy-Accuracy-Security Trade-offs in Resistive In-memory Computing Architectures”, **SRC JUMP 2.0 COCOSYS Annual Review**, at Georgia Tech, Atlanta, Georgia, USA, 2025.
- “Energy-Accuracy Trade-Offs for Resistive In-Memory Computing Architectures”, **JxCDC SCS Open Journal Webinar**, online, 2025.
- “Security Vulnerabilities of eNVM-based In-memory Architectures”, **COCOSYS**, online, 2024.
- “Energy-accuracy trade-offs of resistive in-memory computing architectures”, at **IIT Madras**, Chennai, India, 2024.
- “Compute SNDR-boosted 22nm MRAM IMC using Statistical Error Compensation”, **COCOSYS**, online, 2024.
- “Energy-Accuracy Trade-Offs for Resistive In-Memory Computing Architectures”, **COCOSYS**, online, 2024.
- “GPLeak: Software Power Side Channel Leakage in GPUs”, **CSL Student Conference, Security and Privacy Session (Oral Presentation Award)**, at UIUC, Urbana, Illinois, USA, February 2024.
- “Compute SNR-boosted 22nm MRAM-based In-memory Computing Macro using Statistical Error Compensation”, at **ESSCIRC**, Lisbon, Portugal, 2023.
- “Fundamental Limits on the Compute SNR of Resistive In-memory Architectures”, at **CSL Student Conference, Circuits and Systems Session (Best Speaker Award)**, UIUC, Urbana, Illinois, USA, 2023.
- “Fundamental Limits on the Computational Accuracy of Resistive Crossbar-based In-memory Architectures”, at **ISCAS**, Austin, Texas, USA, 2022.

INTERNSHIP EXPERIENCE

Texas Instruments, Kilby Labs

Dallas, USA

Supervised by: Dr. Mahesh Mehendale and Dr. Avishek Biswas

Summer 2021

- **NAND Flash-based IMC Design:** Characterized NAND flash cell properties and designed sensing circuits for IMC operation.
- **Behavioral Modeling of ADC in PyTorch:** Analyzed the impact of ADC non-idealities on the DNN accuracy of IMC implementations using PyTorch models.

Technische Universität Berlin (TU Berlin)

Berlin, Germany

Supervised by: Prof. Dr.-Ing. Friedel Gerfers

Summer 2017

- Supported under the Deutscher Akademischer Austauschdienst Working Internships in Science and Engineering (DAAD WISE) fellowship program.
- **SAR-ADC Digital Block Optimization:** Reduced power consumption of a 10-bit, 100 MS/s SAR ADC digital block from 1.7 mW to 1.1 mW in 65nm TSMC technology through circuit- and logic-level optimizations.
- **Low-Power Flip-Flop Study:** Evaluated flip-flop architectures for power-delay trade-offs; adopted transmission gate-based pulsed flip-flops achieving 50% power reduction over conventional designs.

Semiconductor Laboratory, Dept. of Space, Govt. of India

Chandigarh, India

Supervised by: H. S. Jattana

Summer 2016

- **Two-Stage Op-Amp Design:** Designed and characterized a two-stage op-amp (73 dB gain, 50 MHz bandwidth, 55° phase margin, 20 V/ μ s slew rate) in 180nm CMOS.
- **Clocked Comparator Design:** Designed a latch-based clocked comparator (50.37 dB gain, 4.7 ns delay) at 100 MHz with 2.5 mW power consumption.

TEACHING AND MENTORSHIP EXPERIENCE

Northeastern University

Fall 2025

EECE 5699: Computer Hardware and System Security, Co-Course Instructor

- **Primary Instructor:** Prof. Yunsi Fei
- Assisted in preparing homework and delivering lectures.

University of Illinois at Urbana-Champaign

Fall 2022

ECE 342: Electronics Circuits, Teaching Assistant

- **Instructors:** Prof. Pavan Hanumolu and Prof. Chandrasekhar Radhakrishnan
- Helped prepare homework and exams, held office hours, and assisted students in understanding course material and solving problems.

Mentorship Experience

- **Han-Mo Ou:** PhD Student, University of Illinois at Urbana-Champaign
- **Kaining Zhou:** PhD Student, University of Illinois at Urbana-Champaign
- **Yufei Wang:** PhD Student, Northeastern University
- **Tianhong Xu:** PhD Student, Northeastern University
- **Thomas Gourousis:** PhD Student, Northeastern University

RESEARCH EXPERIENCE

- **Temperature Sensor-based On-chip Trojan Detection** (under review)
 - First experimental validation of hardware Trojan detection via on-chip differential temperature sensors on a fabricated 65-nm CMOS mixed-signal ASIC.
 - Achieved 95% detection accuracy by exploiting electrothermal coupling to capture Trojan-induced thermal perturbations without circuit modifications.
 - Outperformed power side-channel baseline with $6.65\times$ higher detection SNR and 25% improved accuracy across 50 kHz–8 MHz range of operating frequency.
- **Chip-specific Backdoor Attacks on Diffusion Transformer Accelerators** (under submission)
 - Introduced the first chip-specific backdoor attack for DiT accelerators, enabling selective targeting of individual chips or chip families based on unique hardware signatures.
 - Developed two triggering mechanisms: architecture triggers exploiting 4-bit/8-bit MVM mode-switching, and circuit triggers derived from SRAM power-up states due to process variations.
 - Demonstrated on a 28nm digital in-memory computing prototype with >99.9% selectivity on targeted chips and undetectable model quality (FID<15).
- **Inference-time Backdoor Attacks via Memory Faults** (ICCAD'26 (to appear))
 - Proposed a Rowhammer-based backdoor attack that bridges the gap between algorithmic design and hardware realization by accounting for device-specific bit-flip patterns.
 - Developed deterministic DRAM vulnerability profiling and a DRAM page selection algorithm that maximizes attack success rate while constraining test accuracy degradation.
 - Achieved ~90% attack success rate and >83% test accuracy across three DDR4 chips for FP32 and INT8 models, requiring 64–74% fewer DRAM pages than prior work.
- **Unsupervised Deep Learning based Side-Channel Attacks** (AsiaCCS'26)
 - Proposed a deep-learning-based unsupervised side-channel attack trained with unlabeled target traces to predict a secret cryptographic key, across different hardware platforms and side-channel modalities.
 - Demonstrated transferability of the trained model with labeled traces in a source domain (power traces from Microcontroller) to the target domain with unlabeled traces (EM traces from FPGA).

- **Security Vulnerabilities of Resistive IMCs** (ICCAD'24, IEDM'24, TCAD'26)
 - Proposed a statistical framework for model extraction attacks (MEAs) on resistive IMCs, and employed it to construct three query-based attacks: basis vector (BV), least squares (LS), and stochastic gradient descent (SGD) attacks.
 - Attack on real-life MRAM-based IMC prototype revealed that it is indeed vulnerable to MEAs despite the presence of analog non-idealities.
 - Energy-accuracy-security trade-offs for eNVM-based IMCs reveal that they can simultaneously achieve high energy efficiency, inference accuracy, and enhanced resilience against MEAs by operating at low compute accuracy.
- **Energy-Accuracy Trade-offs in Resistive Parallel-bars** (JxCDC'24)
 - Formulated behavioral models for circuit non-idealities at the array, sensing, and ADC levels and verified with chip measurements.
 - Evaluated the limits of compute-SNDR and its dependencies on device, circuit, and architecture parameters, obtaining a maximum SNDR of 18 dB-to-23 dB, barely meeting the requirement for achieving network-level accuracy close to digital.
 - Highlights the need for SNDR-boosting methods, such as statistical error compensation (SEC) and noise-aware training.
- **Algorithmic Error Compensation for MRAM IMC** (ESSCIRC'23, JSSC'24)
 - Characterized the compute SNDR and the energy-accuracy trade-offs for a 22 nm MRAM IMC chip.
 - Proposed SNDR-boosting techniques, including offset-compensated current sensing (OCCS) to minimize static ADC column mismatch and algorithmic statistical error correction (SEC) to reduce wire parasitic induced non-linearity.
 - Obtained a $5 \times$ reduction in energy by trading-off SEC-enabled SNDR gain of 2.6 dB-to-6 dB.
 - Demonstrated SEC-enabled increase in neural network (NN) accuracy from 74.8% to 82.0% for CIFAR-10 over ResNet-20 without resorting to noise-aware training.
- **Accuracy Limits of Resistive Crossbars** (ISCAS'22)
 - Evaluated the fundamental limits on compute-SNDR for MRAM, ReRAM, and FeFET-based crossbars using behavioral models verified in 22 nm commercial node.
 - Analyzed the dependence of compute SNDR on sensing resistance R_s , ADC precision B_{ADC} (circuit), dot-product dimension N (architecture), and the resistive contrast $R_{\text{off}}/R_{\text{on}}$ (device).
 - Increasing device level resistive contrast provides diminishing returns in SNDR since the input DAC and conductance mismatch begin to dominate.
 - Demonstrated that network-level accuracy correlates with the bank-level SNDR, revealing the accuracy-maximizing parameters using SNDR analysis.
- **Benchmarking ML Accelerators** (OJ-SSCS'22, CICC'22)
 - Proposed a comprehensive benchmarking methodology leveraging a compositional view.
 - Benchmarked > 70 ML IC designs published in major circuit conferences (ISSCC, VLSI, CICC) since 2018 to comprehend the limits and identify challenges in ML accelerator design space.
 - SRAM-based IMCs have higher energy efficiency (TOPS/W) and compute density (TOPS/mm²) over digital accelerators at the bank level, but the gap reduces dramatically at the processor level.
 - Resistive IMCs currently lag behind SRAM IMCs and digital accelerators in both TOPS/W and TOPS/mm² due to their low compute accuracy.

TECHNICAL SKILLS

Programming	Python (PyTorch), Verilog, VerilogA, MATLAB, C
EDA Tools	Cadence Virtuoso, Genus, Innovus, SPICE, Xilinx Vivado
ASIC Testing	KiCAD (PCB design), worked with PYNQ, Arduino, and Atmel AVR Testing Boards

GRADUATE COURSEWORK

Circuits: Digital IC Design, Analog IC Design

Signal Processing: Advanced DSP, Vector Space Signal Processing

Systems: Random Processes, Detection and Estimation Theory, Machine Learning

Computer Science: Computer Architecture, Advanced Computer Security, Hardware Security

PROFESSIONAL ACTIVITIES AND SERVICES

- **Technical Program Committee:** ISCA'26, MICRO'26, ICCAD'26
- **Journal Reviewer:** IEEE JSSC, IEEE TVLSI, IEEE TCAD, IEEE TCAS-I, IEEE TCAS-II, IEEE JETCAS, IEEE TACO, IEEE TED, IEEE D&T
- **Conference (External) Reviewer:** IEEE ISCAS, ACM CCS, ACM DAC, NDSS.

REFERENCES

Prof. Naresh Shanbhag

Professor, Department of ECE,
University of Illinois at Urbana-Champaign
shanbhag@illinois.edu

Prof. Pavan Hanumolu

Professor, Department of ECE,
University of Illinois at Urbana-Champaign
hanumolu@illinois.edu

Prof. Yunsi Fei

Professor, Department of ECE,
Northeastern University
y.fei@northeastern.edu